what the dataset provides
genomic DNA
5' UTR
CDS
3' UTR
protein
protein held constant
multiple species
tissues or cell types
in hand
research plan priority
mRNA stability (CodonBERT / Diez 2022)
1
mRFP synonymous variants (CodonBERT)
5, parallel
Fungal expression (Wint 2022)
1
E. coli expression (Ding 2022)
1
same coding sequence
Ab1 antibody mRNA expression (Yazdani)
1, added
CoV-Vac mRNA degradation (Leppek 2022)
1, added
GTEx isoform expression (IsoFormer)
2
mRNA half-life, Saluki (mRNABench)
3
LBKWK synthetic mRNA: half-life and MRL
3, 4, added
Translation efficiency (mRNABench)
4
distinct molecules
Mean ribosome load, Sugimoto (mRNABench)
4, added
COMET multi-molecule: RPI, EPI, CRISPR off-target
listed
Protein-to-mRNA ratio (Eraslan)
CaLM suite: abundance, Tm, solubility
COMET cross-molecule CDS/protein: Flu, EC, Beta-Lac
listed
codon vs amino acid
mRNABench: GO, localization, RBP, UTRs
5' UTR ribosome load (MRL MPRA)
3' UTR polyadenylation (APA)
single region, no protein
Enhancer perturbation (CDT)
regulatory DNA
filled: available
hatched: partly (see text)
grey: not available
listed: from the October paper list